

```
In[*]:= Clear[pathHome, pathFGV, path]
pathHome = "G:\\Meu Drive\\ArtigoWikipedia\\PesquisaWikipedia\\";
pathFGV =
  "D:\\Users\\leandro.gorno\\Dropbox\\Research\\Projects\\2023-11 - Wikipedia
  graphs\\2025-07 - New data gathering\\Data\\";
path = pathHome;
```

```
In[*]:= Clear[graphPT3]
graphPT3 = Get[path <> "graphPT3.dat"]
```

Out[*]=

Graph [ Vertex count: 25092
Edge count: 55474]

```
In[*]:= (*1. Define your 8 authors*)
targets = {"Carolina Maria de Jesus", "Clarice Lispector", "Gabriela Mistral",
  "Júlia Lopes de Almeida", "Machado de Assis",
  "Lima Barreto (escritor)", "Mário de Andrade", "Jorge Luis Borges"}
cleangraph = graphPT3;
```

```
(*2. Find all edges connected to ANY of the 8 targets*)
targetEdges = Select[EdgeList[cleangraph], IntersectingQ[List@@#, targets] &];
```

```
(*3. Extract all unique vertices involved (The 8 Egos+their network*)
combinedVertices = Union[Flatten[List@@@targetEdges]];
```

```
(*4. Create the master combined graph!*)
GTOTAL = combinedGraph = Subgraph[cleangraph, combinedVertices];
```

```
Print["Nodes in combined network: ", VertexCount[combinedGraph]];
Print["Edges in combined network: ", EdgeCount[combinedGraph]];
```

Out[*]=

{Carolina Maria de Jesus, Clarice Lispector, Gabriela Mistral, Júlia Lopes de Almeida,
Machado de Assis, Lima Barreto (escritor), Mário de Andrade, Jorge Luis Borges}

Nodes in combined network: 888

Edges in combined network: 3589

```

In[*]:= (*Helper function-define ONCE,at top level*)ClearAll[importWithRetry];
importWithRetry[url_String] :=
Module[{data, tries = 0, pause = 1.0, maxRetries = 4}, data = $Failed;
While[(data == $Failed || ! AssociationQ[data]) && tries < maxRetries,
If[tries > 0, Pause[pause];
pause *= 2];
data = Quiet@Check[Import[url, "RawJSON"], $Failed];
tries++];
data];

(*Main function*)
ClearAll[getGenero];
getGenero[titulo_String] := Module[{apiWiki, dadosWiki, pagina, qid, apiWD,
dadosWD, claims, generoQ, apiLabel, dadosLabel, result}, Pause[1.0];
(*Step 1:Wikipedia pageprops*)apiWiki =
"https://pt.wikipedia.org/w/api.php?action=query&prop=pageprops&titles=" <>
URLEncode[StringReplace[titulo, "_" -> " "]] <> "&format=json";
dadosWiki = importWithRetry[apiWiki];
If[! AssociationQ[dadosWiki], Return["none"]];
pagina = Quiet@Check[First@Values@dadosWiki["query", "pages"], $Failed];
If[! AssociationQ[pagina], Return["none"]];
qid = pagina["pageprops", "wikibase_item"];
If[MissingQ[qid] || ! StringQ[qid], Return["none"]];
(*Step 2:Wikidata claims*)
apiWD = "https://www.wikidata.org/w/api.php?action=wbgetentities&ids=" <>
qid <> "&props=claims&format=json";
dadosWD = importWithRetry[apiWD];
If[! AssociationQ[dadosWD], Return["none"]];
claims = Quiet@Check[dadosWD["entities", qid, "claims"], $Failed];
If[! AssociationQ[claims], Return["none"]];
If[! KeyExistsQ[claims, "P21"], Return["none"]];
generoQ =
Quiet@Check[claims["P21"][[1, "mainsnak", "datavalue", "value", "id"]], $Failed];
If[! StringQ[generoQ], Return["none"]];
(*Step 3:Label*)
apiLabel = "https://www.wikidata.org/w/api.php?action=wbgetentities&ids=" <>
generoQ <> "&props=labels&languages=pt&format=json";
dadosLabel = importWithRetry[apiLabel];
If[! AssociationQ[dadosLabel], Return["none"]];
result =
Quiet@Check[dadosLabel["entities", generoQ, "labels", "pt", "value"], $Failed];
If[StringQ[result], ToLowerCase[result], "none"]];

```

```

In[*]:= (* =====
CELL A:Fetch genders only for labeled nodes (with disk
        cache)=====*)
(*---A.0.Cache file location---*)
genderCacheFile = FileNameJoin[{path, "genderCache.m"}];

(*---A.1.Determine the set of labeled nodes---*)
gVertices = VertexList[GTOTAL];
cleanDegrees = AssociationThread[VertexList[cleangraph] → VertexDegree[cleangraph]];
gVertexDegrees = AssociationMap[cleanDegrees[#] &, gVertices];
top200 = Keys[TakeLargest[gVertexDegrees, UpTo[200]]];

communities = FindGraphCommunities[GTOTAL, Method → "Modularity"];
representatives = Table[Module[{sub, ec, assoc}, sub = Subgraph[GTOTAL, comm];
  If[VertexCount[sub] ≤ 1, comm, ec = Quiet@EigenvectorCentrality[sub];
  If[! ListQ[ec] || Length[ec] != VertexCount[sub], ec = VertexDegree[sub]];
  assoc = AssociationThread[VertexList[sub] → ec];
  Keys[TakeLargest[assoc, UpTo[5]]]], {comm, communities}];
allRepresentatives = Union @@ representatives;

labeledVertices = Union[top200, targets, allRepresentatives];

(*---A.2.Load cache from disk if available,else initialize---*)
If[FileExistsQ[genderCacheFile], genderCache = Get[genderCacheFile];
  Print["Loaded cache from disk: ", Length[genderCache], " entries."],
  genderCache = <|>;
  Print["Initialized empty cache."]];

(*---A.3.Fetch only what's missing or previously failed---*)
nodesToQuery =
  Select[labeledVertices, ! KeyExistsQ[genderCache, #] || genderCache[#] == "none" &];

Print["Querying Wikidata for ", Length[nodesToQuery], " nodes..."];

Monitor[Do[genderCache[nodesToQuery[[k]]] = getGenero[nodesToQuery[[k]]],
  {k, Length[nodesToQuery]}], Row[{"Progress: ", k, "/", Length[nodesToQuery]}]];

(*---A.4.Save cache to disk---*)
Put[genderCache, genderCacheFile];
Print["Saved cache to: ", genderCacheFile];
Print["Final distribution: ", Counts[Values[genderCache]]];

Loaded cache from disk: 458 entries.

Querying Wikidata for 0 nodes...

Saved cache to: G:\Meu Drive\ArtigoWikipedia\PesquisaWikipedia\genderCache.m
Final distribution: <|feminino → 107, masculino → 348, none → 2, mulher trans → 1|>

```

```
In[*]:= allRepresentatives
```

```
Out[*]=
```

```
{Afrânio Peixoto, Anita Malfatti, Anne Chapman, Antônio Carlos Secchin, Artur Azevedo,
Audálio Dantas, Augusto Meyer, Baruch Espinoza, Carolina Maria de Jesus,
Chico Buarque, Clarice Lispector, Edu Lobo, Fernando Sabino, Fiódor Dostoiévski,
Francisco de Assis Barbosa, James Joyce, Jarid Arraes, Jonathan Swift,
Jorge Luis Borges, José de Alencar, José Guilherme Merquior, Luciano de Samósata,
Machado de Assis, Marcos Almir Madeira, Mariana Crioula, Mário de Andrade,
Menotti Del Picchia, Olavo Bilac, Oswald de Andrade, Poeta Margarida,
Sérgio Paulo Rouanet, Tarsila do Amaral, Tia Ciata, Victor Hugo, Vinicius de Moraes}
```

```
In[*]:= getGenero["Victoria Ocampo"] (*should return "masculino"*)
getGenero["Clarice Lispector"] (*should return "feminino"*)
```

```
Out[*]=
```

```
feminino
```

```
Out[*]=
```

```
feminino
```

```
In[*]:= (* =====
Update Gender Cache for the new Top 25%
Nodes=====*)
```

```
(*1. Ensure cache file location is set*)
```

```
genderCacheFile = FileNameJoin[{path, "genderCache.m"}];
```

```
(*2. Load existing cache so we don't start from scratch*)
```

```
If[FileExistsQ[genderCacheFile], genderCache = Get[genderCacheFile];
```

```
Print["Loaded cache with ", Length[genderCache], " existing entries."],
```

```
genderCache = <| |>;
```

```
Print["Started new cache."]]];
```

```
(*3. Find exactly who is missing from the NEW list of labeled vertices*)
```

```
nodesToQuery =
```

```
Select[labeledVertices, ! KeyExistsQ[genderCache, #] || genderCache[#] == "none" &];
```

```
Print["Querying Wikidata for ", Length[nodesToQuery], " missing nodes..."];
```

```
(*4. Fetch the missing genders (with a progress bar)*)
```

```
Monitor[Do[genderCache[nodesToQuery[[k]]] = getGenero[nodesToQuery[[k]]],
```

```
{k, Length[nodesToQuery]}],
```

```
Row[{"Fetching: ", k, " / ", Length[nodesToQuery], " (", nodesToQuery[[k]], ")"}]]];
```

```
(*5. Save the updated cache back to your hard drive*)
```

```
Put[genderCache, genderCacheFile];
```

```
Print["Done! Saved updated cache to disk."];
```

```
Loaded cache with 458 existing entries.
```

```
Querying Wikidata for 4 missing nodes...
```

```
Done! Saved updated cache to disk.
```

```
In[*]:= Counts[Values[genderCache]]
```

```
Out[*]=
```

```
<|feminino → 107, masculino → 348, none → 2, mulher trans → 1|>
```

```
(*4. Fetch the missing genders WITH A PAUSE to prevent API blocking*)
```

```
Monitor[Do[Module[{result},
```

```
  (*Ask Wikidata for the gender*)result = getGenero[nodesToQuery[[k]]];
```

```
  (*Save the result to the cache*)genderCache[nodesToQuery[[k]]] = result;
```

```
  (*Pause for half a second before asking the next one.This stops Wikidata from  
  blocking your IP address!*)Pause[0.5];], {k, Length[nodesToQuery]}],
```

```
Row[{"Fetching: ", k, " / ", Length[nodesToQuery], " (", nodesToQuery[[k]], ")"}]]];
```

```
(* =====
```

```
Community plot with global degree filtering,
```

```
target highlighting,community representatives,full-canvas labels,
```

```
and GENDERED LABELS=====*)
```

```
(*---0. Inputs assumed to exist:cleangraph,G,targets,genderCache---*)
```

```
(*---1. Filter Graph:Keep Top 25% (Cut the bottom 75%)---*)
```

```
originalGVertices = VertexList[GTOTAL];
```

```
(*Safely calculate 0.6 In+0.4 Out using the ORIGINAL cleangraph*)
```

```
rawCentrality = If[DirectedGraphQ[cleangraph], AssociationMap[
```

```
  (0.6 * VertexInDegree[cleangraph, #] + 0.4 * VertexOutDegree[cleangraph, #]) &,  
  originalGVertices],
```

```
AssociationMap[VertexDegree[cleangraph, #] &, originalGVertices]]];
```

```
(*Find the 50th percentile cutoff line*)
```

```
cutoffValue = Quantile[Values[rawCentrality], 0.50];
```

```
(*Keep nodes above the cutoff,BUT always ensure your targets are kept*)
```

```
topNodes = Select[Keys[rawCentrality], rawCentrality[#] ≥ cutoffValue &];
```

```
nodesToKeep = Union[topNodes, targets];
```

```
(*OVERWRITE G with the new,cleaner subgraph.The rest of
```

```
the pipeline will now use this lighter graph automatically.*)
```

```
G = Subgraph[GTOTAL, nodesToKeep];
```

```
(*Recalculate variables for the new,smaller G*)
```

```
(*labeled vertices esta definido dos veces*)
```

```
gVertices = VertexList[G];
```

```
labeledVertices = Union[gVertices, targets];
```

```
(*---2. Communities (or use your existing list)---*)
```

```
communities = FindGraphCommunities[G, Method → "Modularity"];
```

```
n = Length[communities];
```

```
(*---3. Top 5 representative nodes per community---*)
```

```
representatives = Table[Module[{sub, ec, assoc}, sub = Subgraph[G, comm];
```

```
  If[VertexCount[sub] ≤ 1, comm, ec = Quiet@EigenvectorCentrality[sub];
```

```
  If[! ListQ[ec] || Length[ec] != VertexCount[sub], ec = VertexDegree[sub]]];
```

```
  assoc = AssociationThread[VertexList[sub] → ec];
```

```
  Keys[TakeLargest[assoc, UpTo[5]]]]], {comm, communities}];
```

```

allRepresentatives = Union @@ representatives;
(*Aca tambien esta definido dos veces allrepresentatives*)

(*---4. Per-community layouts (rescaled)---*)
subgraphs = Subgraph[G, #] & /@ communities;
localLayouts = GraphEmbedding[#, "SpringElectricalEmbedding"] & /@ subgraphs;
localLayouts = (# - ConstantArray[Mean[#], Length[#]]) & /@ localLayouts;

(*Circularize: rescale each layout so its x-extent and y-extent match*)
localLayouts = Map[Function[pts, If[Length[pts] ≤ 1, pts,
  Module[{xs = pts[[All, 1]], ys = pts[[All, 2]], rx = Max[Abs[xs]],
    ry = Max[Abs[ys]]};
  If[rx < 10^-6 || ry < 10^-6, pts, s = Sqrt[rx * ry];
    Transpose[{xs * (s / rx), ys * (s / ry)}]]]], localLayouts];

(*Inflate radii to account for node size+label space.*)
nodeSizePad = 2.0;
labelPad = 4.0;
extraGap = 12.0; (*Increased massively to fight the gravity of the new hubs*)

rawRadii =
  MapThread[If[Length[#1] ≤ 1, 0.5, Max[Norm /@ #2]] &, {communities, localLayouts}];
radii = rawRadii + nodeSizePad + labelPad;

(*---5. Meta-graph layout, then iterative non-overlap relaxation---*)
vertexToComm =
  Association[Flatten[MapIndexed[Thread[#1 → First[#2]] &, communities]]];
interEdges =
  Cases[EdgeList[G], (a_ ↔ b_ | a_ → b_) /; vertexToComm[a] != vertexToComm[b] →
    UndirectedEdge[vertexToComm[a], vertexToComm[b]]];
edgeWeights = Counts[Sort /@ interEdges];
metaG = Graph[Range[n], Keys[edgeWeights], EdgeWeight → Values[edgeWeights]];
metaCoords = If[n == 1, {{0., 0.}}, GraphEmbedding[metaG, "SpringElectricalEmbedding"]];

(*Initial uniform scale*)
If[n > 1, minMetaDist =
  Min[Table[EuclideanDistance[metaCoords[[i]], metaCoords[[j]], {i, n}, {j, i + 1, n}]];
  requiredMin = 2 Max[radii] + extraGap;
  metaCoords = metaCoords * (requiredMin / Max[minMetaDist, 0.001]);];

(*Iterative pairwise relaxation*)
relaxCommunitiesByPoints[centers_, localPts_, gap_, maxIter_ : 300] :=
  Module[{c = centers, k = Length[centers], moved, iter = 0, globalPts,
    dir, push, minSep, dvec, dij}, While[iter < maxIter, moved = False;
    globalPts = MapThread[Function[{pts, ctr}, (# + ctr) & /@ pts], {localPts, c}];
    Do[minSep = Min[Outer[EuclideanDistance, globalPts[[i]], globalPts[[j]], 1]];
    If[minSep < gap, dvec = c[[j]] - c[[i]];
      dij = Norm[dvec];
      dir = If[dij < 10^-6, RandomReal[{-1, 1}, 2], dvec / dij];
      push = 0.5 (gap - minSep);

```

```

      c[[i]] -= push dir;
      c[[j]] += push dir;
      moved = True;], {i, k}, {j, i + 1, k}];
    If[! moved, Break[]];
    iter++;];
  c];

metaCoords = If[n > 1,
  relaxCommunitiesByPoints[metaCoords, localLayouts, extraGap + labelPad], metaCoords];

globalCoords = Flatten[MapThread[
  Function[{verts, localPts, center}, Thread[verts → (# + center & /@ localPts)]],
  {communities, localLayouts, metaCoords}], 1];

(*---6. Build vertex labels and styles---*)
colorFemale = RGBColor["#BA4A34"]; (*Darker Terracotta*)
colorMale = RGBColor["#005959"]; (*Darker Teal*)
colorTrans = RGBColor["#8A2BE2"]; (*Violet (unchanged)*)
colorUnknown = RGBColor["#555555"]; (*Slightly darker gray*)

(*... the rest of the safeGet and
  getGenderColor functions remain exactly the same...*)

(*1. Aggressive lookup to bypass String vs.Symbol mismatches*)
safeGet[dict_, key_] :=
  If[KeyExistsQ[dict, key], dict[key], Module[{strKey = ToString[key], matches},
    matches = Select[Keys[dict], ToString[#] == strKey &];
    If[Length[matches] > 0, dict[First[matches]], "none"]]];

(*2. Extract and match the Portuguese gender strings*)
getGenderColor[v_] := Module[{val, gStr}, val = safeGet[genderCache, v];
  (*Unwrap Mathematica Entity objects if present*)
  If[Head[val] === Entity, val = val[[2]]];
  (*If it is a list, flatten it and take the first item*)
  If[ListQ[val], val = If[Length[val] > 0, First[Flatten[val]], "none"]];
  (*Convert to string, lowercase, and remove ALL spaces*)
  gStr = StringReplace[ToLowerCase[ToString[val]], Whitespace → ""];
  (*Match against the PORTUGUESE text*)
  Switch[gStr, "feminino", colorFemale, "masculino", colorMale, "trans",
    colorTrans, "transgênero", colorTrans, "mulhertrans", colorTrans, "homemtrans",
    colorTrans, "q6581072", colorFemale, (*Wikidata code for Female*)
    "q6581097", colorMale, (*Wikidata code for Male*)_ , colorUnknown]];

(*3. Apply styles*)
vertexLabelRules =
  Map[Function[v, v → Which[MemberQ[target, v], Placed[Style[v, Bold, 18,
    getGenderColor[v]], Above], MemberQ[allRepresentatives, v],
    Placed[Style[v, Bold, 17, getGenderColor[v]], Above], MemberQ[labeledVertices, v],
    Placed[Style[v, 11, getGenderColor[v]], Above], True, None]], VertexList[G]];

```

```

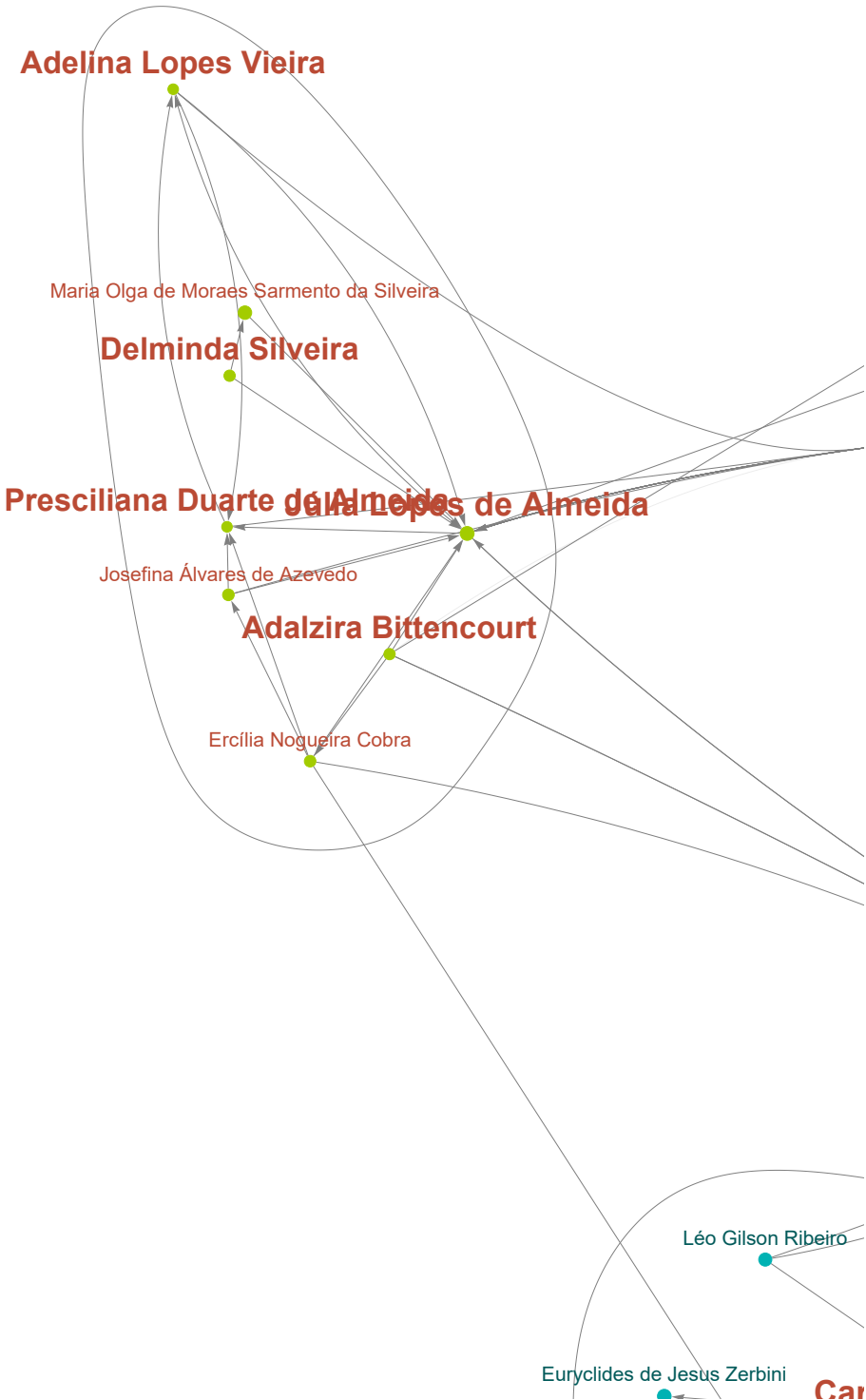
(*---Calculate Custom Centrality for Sizing---*) (*Safely calculate 0.6 In+
0.4 Out using the ORIGINAL cleangraph to reflect global importance*)
rawCentrality = If[DirectedGraphQ[cleangraph], AssociationMap[
  (0.6 * VertexInDegree[cleangraph, #] + 0.4 * VertexOutDegree[cleangraph, #]) &,
  VertexList[G]], AssociationMap[VertexDegree[cleangraph, #] &, VertexList[G]]];

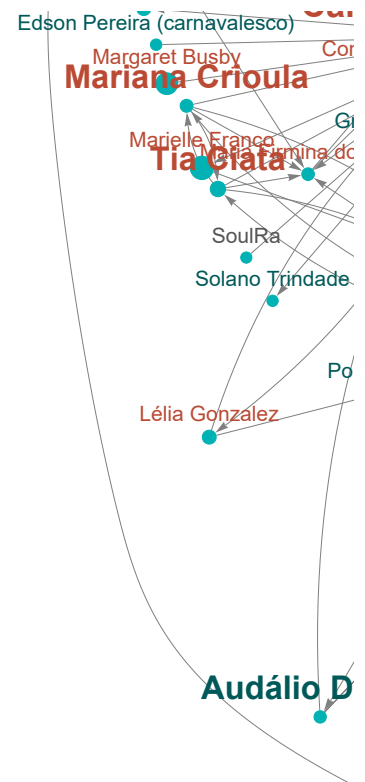
minCent = Min[Values[rawCentrality]];
maxCent = Max[Values[rawCentrality]];
(*Strictly continuous sizing (no categories). We use the Square Root of the
centrality to gently flatten the power-law curve. The scale is anchored
between 0.4 (small but visible) and 2.5 (large but not massive).*)
vertexSizeRules = Map[Function[v, v → Rescale[Sqrt[rawCentrality[v]],
  {Sqrt[minCent], Sqrt[maxCent]}, {0.6, 4.0}]], VertexList[G]];
(*---7. Bake coordinates and styling into the graph---*)
Gpos = Graph[G, VertexCoordinates → globalCoords,
  VertexLabels → vertexLabelRules, VertexSize → vertexSizeRules,
  VertexLabelStyle → Directive[FontFamily → "Helvetica"]];

(*---8. Plot with generous canvas and no clipping---*)
CP = CommunityGraphPlot[Gpos, communities, Method → {"CommunityRegionStyle" → "Disk"},
  ImageSize → {2400, 2400}, PlotRangePadding → Scaled[0.08], AspectRatio → 1,
  EdgeStyle → Directive[Opacity[0.15], Gray, Arrowheads[0.003]], PlotTheme → "Detailed"]

```

Out[]=





```
In[ ]:= allRepresentatives
```

Out[*]=

```
{Adalzira Bittencourt, Adelina Lopes Vieira, Afrânio Peixoto, Ana Beatriz Nogueira,
Anita Malfatti, Arthur Schopenhauer, Artur Azevedo, Audálio Dantas,
Baruch Espinoza, Carolina Maria de Jesus, Chico Buarque, Delminda Silveira,
Edu Lobo, Eliane Giardini, Fernando Sabino, Hélio Pellegrino, Jarid Arraes,
Jorge Luis Borges, Júlia Lopes de Almeida, Machado de Assis, Manoel Carlos,
Marcos Caruso, Maria Bethânia, Mariana Crioula, Mário de Andrade, Menotti Del Picchia,
Olavo Bilac, Oswald de Andrade, Otto Lara Resende, Paulo Mendes Campos,
Pitágoras, Platão, Presciliana Duarte de Almeida, Ruy Barbosa, Sócrates,
Susana Vieira, Tarsila do Amaral, Tia Ciata, Tom Jobim, Vinicius de Moraes}
```

In[*]:= **genderCache**

Out[*]=

```
<|Adalzira Bittencourt → feminino, Adélia Prado → feminino,
Adelina Lopes Vieira → feminino, Agamenon Magalhães → masculino,
Agatha Christie → feminino, Agostinho da Silva → masculino,
Alfredo Bosi → masculino, Álvaro Lins → masculino, Ana Beatriz Nogueira → feminino,
Ana María Martínez Sagi → feminino, Ana Miranda → feminino, André Carneiro → masculino,
Ángel Parra → masculino, Anne Chapman → feminino, Antonio Candido → masculino,
Antônio Correia de Oliveira → masculino, Antonio Fernando de Franceschi → masculino,
Anton Tchekhov → masculino, Aracy Balabanian → feminino, Artur Azevedo → masculino,
Audálio Dantas → masculino, Augusto Frederico Schmidt → masculino,
Baruch Espinoza → masculino, Bertha Worms → feminino, Beth Goulart → feminino,
Betty Milan → feminino, Betty Vidigal → feminino, Carlos Guastavino → masculino,
Carlos Heitor Cony → masculino, Carolina Maria de Jesus → feminino,
Carolina Sá → feminino, Che Guevara → masculino, Chiara Mastroianni → feminino,
Chica Xavier → feminino, Chico César → masculino, Clara Camarão → feminino,
Clarice Lispector → feminino, Clarice Niskier → feminino, Claude Farrère → masculino,
Conceição Evaristo → feminino, Curt Meyer-Clason → masculino,
Daniel Granieri → masculino, Darcy Vargas → feminino, Delminda Silveira → feminino,
Diamela Eltit → feminino, DJ Oliveira → none, Edgar Allan Poe → masculino,
Edgar Duvivier → masculino, Edson Pereira (carnavalesco) → masculino,
Edy Lima → feminino, Elena Ferrante → feminino, Elifas Andreato → masculino,
Elisa Lispector → feminino, Elizabeth Bishop → feminino,
Ercília Nogueira Cobra → feminino, Ézio Macedo Ribeiro → masculino,
Euryclides de Jesus Zerbini → masculino, Fausto Fawcett → masculino,
Fauzi Arap → masculino, Félix Peyrallo Carbajal → masculino,
Fernando Sabino → masculino, Filinto de Almeida → masculino,
Fiódor Dostoiévski → masculino, Francisco Brennand → masculino,
François Mauriac → masculino, Franz Kafka → masculino, Frédéric Mistral → masculino,
Gabriela Mistral → feminino, Gabriele d'Annunzio → masculino,
Getúlio Vargas → masculino, Gracyanne Barbosa → feminino,
Gregory Rabassa → masculino, Gustavo V da Suécia → masculino,
Guy de Maupassant → masculino, Helena Solberg → feminino, Hélène Cixous → feminino,
Hélio Pellegrino → masculino, Henfil → masculino, Henri Bergson → masculino,
Henrik Ibsen → masculino, Henriqueta Lisboa → feminino, Hermann Hesse → masculino,
Ida Vitale → feminino, Ilka Soares → feminino, Jacques Maritain → masculino,
James Joyce → masculino, Jarid Arraes → feminino, Jean-Paul Sartre → masculino,
João Antônio (escritor) → masculino, João Anzanello Carrascoza → masculino,
João Cândido → masculino, João do Rio → masculino, Jonathan Swift → masculino,
Jorge de Lima → masculino, Jorge Luis Borges → masculino,
```

José Antônio Ananias Mota → masculino, José Antonio Garcia (cineasta) → masculino,
 José Castello → masculino, José Condé → masculino, José Correia Leite → masculino,
 Josefina Álvares de Azevedo → feminino, José Martí → masculino,
 José Olympio → masculino, José Tolentino de Mendonça → masculino,
 José Vasconcelos → masculino, Josué de Castro → masculino,
 Juana de Ibarbourou → feminino, Judith Cortesão → feminino,
 Júlia Lopes de Almeida → feminino, Júlio Verne → masculino, Laila Garin → feminino,
 Lélia Gonzalez → feminino, Leo Cunha → masculino, Léo Gilson Ribeiro → masculino,
 Leonardo Netto → masculino, Leonel Brizola → masculino, Leopoldo Nachbin → masculino,
 Leticia Spiller → feminino, Liev Tolstói → masculino, Lila Ripoll → feminino,
 Lobo (artista plástico) → masculino, Lourival Fontes → masculino,
 Lúcio Cardoso → masculino, Lúcio de Mendonça → masculino,
 Luiz Fernando Carvalho → masculino, Lygia Fagundes Telles → feminino,
 Machado de Assis → masculino, Madalena Schwartz → feminino,
 Manuel Bandeira → masculino, Marcélia Cartaxo → feminino, Margaret Busby → feminino,
 Margarida Lopes de Almeida → feminino, Maria Bethânia → feminino,
 Maria Camargo → feminino, Maria Clara Spinelli → mulher trans,
 Maria de Arruda Müller → feminino, Maria Emília Correia → feminino,
 Maria Firmina dos Reis → feminino, Maria Martins (escultora) → feminino,
 Mariana Crioula → feminino, Mariana Lima → feminino,
 Maria Olga de Moraes Sarmento da Silveira → feminino, María Rosa Oliver → feminino,
 Marie Curie → feminino, Marielle Franco → feminino, Marilena Ansaldi → feminino,
 Mário Faustino → masculino, Mário Schenberg → masculino, Marisa Lajolo → feminino,
 Marjorie Estiano → feminino, Martinho Silva → masculino, Meg Cabot → feminino,
 Mel Lisboa → feminino, Michelle Bachelet → feminino, Moacyr Scliar → masculino,
 Nahum Sirotsky → masculino, Oswaldo de Camargo → masculino, Otávio de Faria → masculino,
 Othon Bastos → masculino, Otto Lara Resende → masculino, Pablo Neruda → masculino,
 Paulo Francis → masculino, Paulo Henriques Britto → masculino,
 Paulo Mendes Campos → masculino, Paul Valéry → masculino,
 Pedro Aguirre Cerda → masculino, Platão → masculino, Porfírio da Paz → masculino,
 Presciliana Duarte de Almeida → feminino, Rachel de Queiroz → feminino,
 Raduan Nassar → masculino, Raimundo Magalhães Júnior → masculino,
 Regina Casé → feminino, Rogério Skylab → masculino, Rubem Braga → masculino,
 Rubén Darío → masculino, Sacha Bali → masculino, Samuel Wainer → masculino,
 Sandy → feminino, Simone de Beauvoir → feminino, Sócrates → masculino,
 Solano Trindade → masculino, SoulRa → none, Stefan Zweig → masculino,
 Suman Pokhrel → masculino, Susana Rinaldi → feminino, Susana Vieira → feminino,
 Suzana Amaral → feminino, Tati de Moraes → feminino, Teresa de la Parra → feminino,
 Tia Ciata → feminino, Tom Jobim → masculino, Toninho Horta → masculino,
 Toussaint Gurgel → masculino, Vicente Huidobro → masculino,
 Vinicius de Moraes → masculino, Wagner Moura → masculino,
 Walter Scott → masculino, Wesley Barbosa → masculino, Zezé Polessa → feminino,
 Lima Barreto (escritor) → masculino, Mário de Andrade → masculino,
 Adonias Filho → masculino, Afrânio Coutinho → masculino, Afrânio Peixoto → masculino,
 Alberto de Oliveira → masculino, Alceu Amoroso Lima → masculino,
 Alexei Bueno → masculino, Alfredo d'Escagnolle Taunay → masculino,
 Almeida Júnior → masculino, Álvares de Azevedo → masculino, Anatole France → masculino,
 Anita Malfatti → feminino, Antônio Carlos Secchin → masculino,
 Antônio Lobo Antunes → masculino, Antunes Filho → masculino,
 Araripe Júnior → masculino, Arthur Rimbaud → masculino, Augusto Meyer → masculino,

Bernardo Guimarães → masculino, Candido Portinari → masculino,
 Carl Gustav Jung → masculino, Carlos Drummond de Andrade → masculino,
 Carlos Haraldo Sörensen → masculino, Cassiano Ricardo → masculino,
 Castro Alves → masculino, Charles Dickens → masculino, Chico Buarque → masculino,
 Coelho Neto → masculino, Dalton Trevisan → masculino, Dante Alighieri → masculino,
 Di Cavalcanti → masculino, Domingo Faustino Sarmiento → masculino,
 Eça de Queiroz → masculino, Eduardo Prado → masculino, Edu Lobo → masculino,
 Eliane Giardini → feminino, Érico Veríssimo → masculino, Euclides da Cunha → masculino,
 Eugénio de Andrade → masculino, Francisco de Assis Barbosa → masculino,
 Gerardo Melo Mourão → masculino, Gonçalves Dias → masculino, Graça Aranha → masculino,
 Gregori Warchavchik → masculino, Guilherme de Almeida → masculino,
 Guimarães Rosa → masculino, Gustave Flaubert → masculino,
 Gustavo Capanema → masculino, Haroldo de Campos → masculino,
 Heitor Villa-Lobos → masculino, Honoré de Balzac → masculino,
 Humberto de Campos → masculino, Inglês de Sousa → masculino,
 Jesus → masculino, João Carneiro de Sousa Bandeira → masculino,
 João Ubaldo Ribeiro → masculino, Joaquim Manuel de Macedo → masculino,
 Joaquim Nabuco → masculino, Joaquim Pedro de Andrade → masculino,
 Johann Wolfgang von Goethe → masculino, Jorge Amado → masculino,
 José de Alencar → masculino, José Guilherme Merquior → masculino,
 José Lins do Rego → masculino, José Maria da Silva Paranhos Júnior → masculino,
 José Veríssimo → masculino, Juan Manuel de Rosas → masculino,
 Julio Cortázar → masculino, Lafayette Rodrigues Pereira → masculino,
 Lauro Müller → masculino, Luciano de Samósata → masculino, Luís de Camões → masculino,
 Luís Viana Filho → masculino, Manuel de Araújo Porto-Alegre → masculino,
 Marcos Almir Madeira → masculino, Mario Vargas Llosa → masculino,
 Marques Rebelo → masculino, Menotti Del Picchia → masculino,
 Miguel Reale → masculino, Millôr Fernandes → masculino, Milton Hatoum → masculino,
 Monteiro Lobato → masculino, Murilo Mendes → masculino, Napoleão Bonaparte → masculino,
 Olavo Bilac → masculino, Orestes Barbosa → masculino, Oswald de Andrade → masculino,
 Patrícia Melo → feminino, Paulo Autran → masculino, Paulo César Saraceni → masculino,
 Paulo Emílio (crítico de cinema) → masculino, Paulo Junqueira Duarte → masculino,
 Pedro II do Brasil → masculino, Poeta Margarida → masculino, Raul Pompeia → masculino,
 Roberto Schwarz → masculino, Rubem Fonseca → masculino, Ruy Barbosa → masculino,
 Sérgio Buarque de Holanda → masculino, Sérgio Milliet → masculino,
 Sérgio Paulo Rouanet → masculino, Sílvio Romero → masculino,
 Stella Maria Rodrigues → feminino, Tarsila do Amaral → feminino,
 Victor Brecheret → masculino, Victor Hugo → masculino, Victoria Ocampo → feminino,
 William Shakespeare → masculino, Yolanda Penteado → feminino,
 Zélia Gattai → feminino, Arthur Schopenhauer → masculino, Manoel Carlos → masculino,
 Marcos Caruso → masculino, Pitágoras → masculino, Claude Lévi-Strauss → masculino,
 Aderbal Freire Filho → masculino, Adolfo Lutz → masculino,
 Affonso Romano de Sant'Anna → masculino, Afonso Celso (poeta) → masculino,
 Alberto Faria → masculino, Alberto Manguel → masculino, Alberto Nepomuceno → masculino,
 Alcides Maia → masculino, Aleijadinho → masculino, Alessandra Negrini → feminino,
 Alfredo Pujol → masculino, Alicia Moreau de Justo → feminino,
 Almeida Garrett → masculino, André Rebouças → masculino, Aníbal Machado → masculino,
 Antônio Botto → masculino, Antônio de Alcântara Machado → masculino,
 Antônio Paim Vieira → masculino, Antônio Torres → masculino,
 Artur de Oliveira → masculino, Astrojildo Pereira → masculino,

Auguste Comte → masculino, Aurélia Rubião → feminino, Aurora Venturini → feminino,
 Bandeira Tribuzi → masculino, Basílio da Gama → masculino, Bernardo Élis → masculino,
 Cacá Carvalho → masculino, Caio Fernando Abreu → masculino,
 Camargo Guarnieri → masculino, Capistrano de Abreu → masculino,
 Carlos de Laet → masculino, Carlos Gomes → masculino, Carlos Lyra → masculino,
 Carlos Magalhães de Azeredo → masculino, Carybé → masculino, Ciro dos Anjos → masculino,
 Cláudio Abramo → masculino, Claudio Willer → masculino, Cleyde Yaconis → feminino,
 Constâncio Alves → masculino, Deodoro da Fonseca → masculino,
 Domicio da Gama → masculino, Domingos Olímpio → masculino, Edney Silvestre → masculino,
 Elsa Garcia Galvez → feminino, Elsie Houston → feminino, Émile Zola → masculino,
 Eneida Maria de Souza → feminino, Eritácio Pessoa → masculino,
 Eugênio Gomes → masculino, Fagundes Varela → masculino, Flávio Aguiar → masculino,
 Fontes de Alencar → masculino, França Júnior → masculino,
 Francisco Otaviano → masculino, Gabriel García Márquez → masculino,
 Gilberto Braga → masculino, Gilda de Melo e Sousa → feminino,
 Glaucio Mattoso → masculino, Grande Otelo → masculino,
 Gregório da Fonseca → masculino, Guillermo de Torre → masculino,
 Guimarães Passos → masculino, Gustavo Corção → masculino,
 Gustavo Franco → masculino, Hans-Joachim Koellreutter → masculino,
 Harold Bloom → masculino, Honório Hermeto Carneiro Leão → masculino,
 Horácio → masculino, Hugo Picchi (ator) → masculino,
 Isabel, Princesa Imperial do Brasil → feminino, Ivani Ribeiro → feminino,
 Jean-Baptiste Debret → masculino, João de Sousa Lima → masculino,
 João Gaspar Simões → masculino, João Ribeiro (escritor) → masculino,
 João Silvério Trevisan → masculino, Jorge Coli → masculino,
 José Rubens Siqueira → masculino, José Wasth Rodrigues → masculino,
 Jô Soares → masculino, Josué Montello → masculino, Junqueira Freire → masculino,
 Lasar Segall → masculino, Laurindo Rabelo → masculino, Luís Aranha → masculino,
 Manoel de Barros → masculino, Manuel Antônio de Almeida → masculino,
 Marco Antônio Pâmio → masculino, Marco Lucchesi → masculino,
 Maria Fernanda Cândido → feminino, Maria Julieta Drummond de Andrade → feminino,
 Mariana Enríquez → feminino, Mário de Alencar → masculino, Mário Quintana → masculino,
 Medeiros e Albuquerque → masculino, Miguel Couto → masculino, Múcio Leão → masculino,
 Muniz Barreto → masculino, Myriam Muniz → feminino, Nélida Piñon → feminino,
 Norah Borges → feminino, Novalis → masculino, Oliveira Lima → masculino,
 Oscar Wilde → masculino, Osman Lins → masculino, Osório César → masculino,
 Oswaldo Aranha → masculino, Paulo Leão (cineasta) → masculino,
 Pedro (apóstolo) → masculino, Pedro Luís Pereira de Sousa → masculino,
 Plínio Doyle → masculino, Poty Lazzarotto → masculino, Quintino Bocaiuva → masculino,
 Raimundo Carrero → masculino, Raimundo Correia → masculino,
 Raimundo de Farias Brito → masculino, Raymundo Faoro → masculino,
 Richard Morse → masculino, Rodolfo Amoedo → masculino, Rodolfo Bernardelli → masculino,
 Rodolfo Nanni → masculino, Rodrigo Melo Franco → masculino, Rodrigo Otávio → masculino,
 Ruth Guimarães → feminino, Salvador de Mendonça → masculino,
 Samuel Beckett → masculino, Sidney Chalhoub → masculino, Sigmund Freud → masculino,
 Silva Ramos → masculino, Silviano Santiago → masculino, Silvina Ocampo → feminino,
 Teófilo Dias → masculino, Urbano Duarte de Oliveira → masculino,
 Valentim Magalhães → masculino, Vicente do Rego Monteiro → masculino,
 Vidiadhar Naipaul → masculino, Virgílio → masculino, Vittorio Gobbis → masculino,
 Wilson Figueiredo → masculino, Wilson Martins (crítico literário) → masculino |>

```

In[*]:= Export[path <> "graphgenderPRUNED.png", CP]
Out[*]=
  G:\Meu Drive\ArtigoWikipedia\PesquisaWikipedia\graphgenderPRUNED.png

In[*]:= n
Out[*]=
  8

In[*]:= Export[path <> "graphgender2.png", CP]
Out[*]=
  G:\Meu Drive\ArtigoWikipedia\PesquisaWikipedia\graphgender2.png

In[*]:= Take[Normal[genderCache], UpTo[5]]
Out[*]=
  {Adalzira Bittencourt → feminino,
   Adélia Prado → feminino, Adelina Lopes Vieira → feminino,
   Agamenon Magalhães → masculino, Agatha Christie → feminino}

In[*]:= vertexColorOf[v_] := Which[! MemberQ[labeledVertices, v],
  neutralColor, Lookup[genderCache, v, "none"] === "masculino", tealColor,
  Lookup[genderCache, v, "none"] === "feminino", terracota, True, neutralColor];

In[*]:= (*Find all labeled nodes that are turning gray*)
grayNodes = Select[labeledVertices, getGenderColor[#] === colorUnknown &];

(*Print the first 10 gray nodes and their exact cache values*)
Take[
  AssociationMap[Lookup[genderCache, #, "MISSING FROM CACHE"] &, grayNodes], UpTo[10]]
Out[*]=
  <|Aderbal Freire Filho → MISSING FROM CACHE,
   Adolfo Lutz → MISSING FROM CACHE, Affonso Romano de Sant'Anna → MISSING FROM CACHE,
   Afonso Celso (poeta) → MISSING FROM CACHE,
   Alberto Faria → MISSING FROM CACHE, Alberto Manguel → MISSING FROM CACHE,
   Alberto Nepomuceno → MISSING FROM CACHE, Alcides Maia → MISSING FROM CACHE,
   Aleijadinho → MISSING FROM CACHE, Alessandra Negrini → MISSING FROM CACHE|>

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